EIGSEP Memo 004

Marjum Pass 2026-07: receiver and temperature calibration

EIGSEP Team

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Abstract

How box-air integrations are put on a kelvin scale, what that scale means, and how far to trust it. The calibration is the minimal three-state Y factor with reflection corrections, built for every integration from the noise-source, load and S11 measurements nearest in time and the lab characterization of the RF-switch paths. It is published as the data product `tcal` and applied by `eigsep_data` when data are loaded. This memo states the method, its validation and coverage, the contaminated calibration visits it drops, where the scale fails at low frequency, the bench measurement of the noise-source temperature that sets the scale, and what remains open. Every figure and number is generated by `make_products.py` in this directory (Appendix A). The measurement-model derivation is `eigsep_cal` memo M003; the dataset is memo 001.

1 Summary

- **What a calibrated temperature is.** The source temperature of the antenna path at reference plane P, the switch matrix's LNA port. It includes the antenna, balun, balun–switch coax and switch path; none is removed. It is not a free-space antenna temperature (§2).
- **The product.** `tcal` v0004 (`marjum-2026-07/derived/tcal/v0004/`); `Bundle.calibrated` in `eigsep_data`. It covers 63 % of box-air antenna integrations between 07-17 04:11 and 07-18 02:58 UTC (73,885 of 116,640); everything else is NaN, never extrapolated (§5).
- **The noise-source temperature is measured.** A bench run on the field switch board puts the diode's excess at P at 491–589 K across the band (ENR 32.3–33.1 dB were it behind the pad), inside the specified 30–35 dB and below the 917 K that `tcal` v0003 assumed. Converted to the field receiver it gives $T_{\rm NS} = 489$–600 K over 50–200 MHz, one spectrum for every integration (§7). The measurement is preliminary: one board temperature, no switch-path loss.
- **It reproduces an independent calibration** of the 16/17 July night (Christian's, 2026-09-25) when given the same $T_{\rm NS} = 917$ K: `tcal` v0003, identical but for that value, agrees to a median ratio of 1.0003 per sky visit (§5). It agrees with Christian's S11 chain to 1×10^{-15} in Γ at P (§4.3).
- **Three calibration visits are dropped.** They carry box-air's 1.000 MHz comb EMI for a minute or two (§4.2).
- **The scale fails below about 30 MHz** and is untrustworthy to about 60 MHz, where the reflection corrections are large and treat loss in the balun and coax as mismatch (§6).
- **Against a GSM × HFSS prediction** the data sit 1.06–1.11 times the simulation at 78–150 MHz with the bench $T_{\rm NS}$, against 1.29–1.66 with 917 K (§6, §7).
- **Open:** how the noise source and its path's reflection change with temperature (the field injection drifts through the night), the switch path's loss, receiver noise waves, and the balun's loss (§8).

2 The switch network and its reference planes

Hardware (Figure 1). The switch matrix connects the receiver to one of several sources, and connects the VNA to any of them so that their reflections can be measured. The VNA reaches the matrix through a cable to

the matrix's VNA port. The sources that have an external connector on the matrix are the antenna (through its balun and the balun–switch coax), the 50 Ω ambient load, and the spare port SP1, which ends a long coax in an open or a short. Two further sources sit on the matrix's circuit board and have no connector: the internal open, short and load standards the VNA is calibrated against in the field, and the noise diode behind a 30 dB pad. The receiver is everything from the matrix's LNA port on: the cable to the first LNA stage (LNA 7, then LNA 8), an attenuator and band-pass filter, the radio-over-fibre link, and after it a second attenuator, band-pass filter and two LNAs ahead of the SNAP.

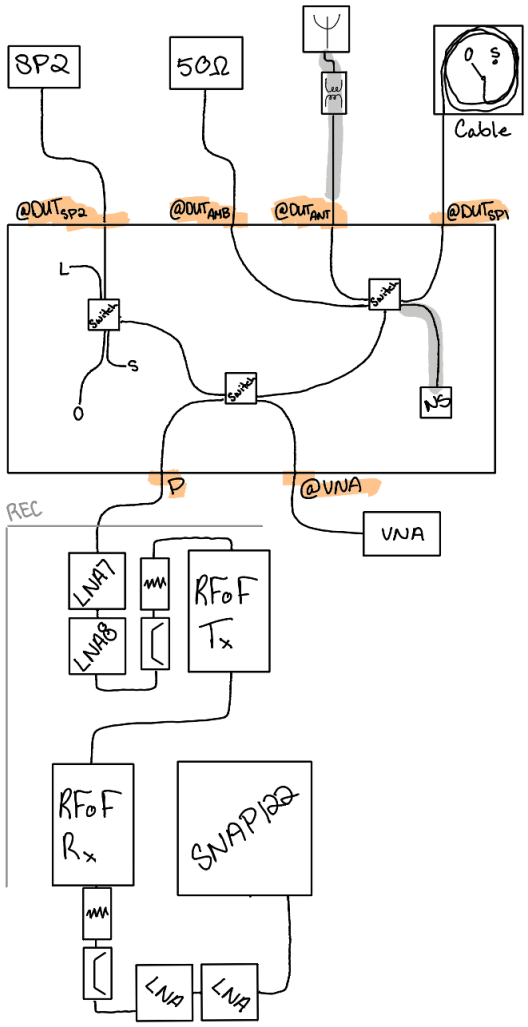


Figure 1: The field system and its reference planes (orange): @VNA, the matrix's VNA port; @DUT, each source's port (SP2, the ambient 50 Ω load, the antenna, SP1); and P, the LNA port. Lines inside the matrix are its switches and connections; each switch path of Table 1 runs along them from one plane to another. The internal open, short and load (O, S, L) and the noise source (NS) sit inside the matrix with no connector. Grey: what has never been characterized, the antenna's balun and balun–switch coax, and the noise source's switch path. The grey line labelled REC marks, loosely, the receiver: everything after P, from the first LNAs (LNA 7, LNA 8) through an attenuator and band-pass filter and the radio-over-fibre (RFoF) link to a second attenuator, band-pass filter and two LNAs ahead of the SNAP.

Reference planes. Three planes are used:

- **@VNA**, the matrix's VNA port: the far end of the VNA cable. The internal standards are characterized

at this plane, and every field sweep is first calibrated to it (§4.3).

- **@DUT**$_i$, the matrix's port for source i: where that source's connector attaches. For the antenna this is the switch end of the balun–switch coax, so $\Gamma_{\rm ant}$ at @DUT includes the antenna, the balun and the coax. Their S-parameters have not been measured, so they cannot be separated from one another. This plane holds each source's reflection without the switch path in front of it.
- **P**, the matrix's LNA port (written @LNA in the lab notes): where the cable to the LNA attaches. The calibration plane cannot be moved further, to the LNA's own input, so the cable belongs to the receiver. P is the plane of every quantity in the calibration (§3): $\Gamma_{\rm rec}$ looks from P into the receiver, and each source's Γ and temperature look from P back through its switch path.

Table 1 lists the switch states, the VNA key each source's sweeps are stored under, and the switch paths that carry its reflection between the planes. Each `VNA*` path joins @VNA to a source's port; each `RF*` path joins P to it. `VNARF` joins @VNA to P directly. The noise diode's paths cannot be characterized: it has no connector to place a standard at. Its reflection at P, which is what the receiver sees, can still be measured in the lab, with the VNA on the matrix's LNA port and the switch on `RFNON` (§7).

state	source behind the switch path	VNA key	@VNA→@DUT	P→@DUT	role
`RFANT`	antenna, balun, balun–switch coax	`ant`	`VNAANT`	`RFANT`	the sky
`RFAMB`	50 Ω ambient load (`tempctrl_load`)	`amb`	`VNAAMB`	`RFAMB`	load reference
`RFSP1_OPEN/SHORT`	long coax, open or short end	`sp1_open/short`	`VNASP1`	`RFSP1`	spare port
SP2	second spare port	—	not characterized		not used
`RFNON`	noise diode on, behind a 30 dB pad	`noise`	not measurable		noise-source ref.
`RFNOFF`	noise diode off, behind the pad	`load`	not measurable		—
receiver	cable to the LNA, and the LNA	`rec`	`VNARF` (@VNA→P)		$\Gamma_{\rm rec}$

Table 1: Switch states and the paths characterized for each. The VNA key `load` is the noise diode *off*; the ambient load is `amb`. Only box-air (correlator input 4 in phase C) is switched; box-gnd cannot be calibrated this way.

Temperatures are calibrated at P, and nothing is de-embedded from a temperature (decision recorded in api.md §5.2): a calibrated antenna temperature includes the antenna, balun, coax and `RFANT` switch path. The @DUT plane is used for reflections only. A comparison with a free-space simulation therefore has to carry the simulation forward through the balun, coax and switch path, not the data back (§6).

3 Measurement model

The three-state Y factor is

$$T^* = T_{\rm load} + T_{\rm NS}\,\frac{P_{\rm ant} - P_{\rm load}}{P_{\rm NS} - P_{\rm load}}, \tag{1}$$

with $T_{\rm load}$ the load's physical temperature and $T_{\rm NS}$ the noise-source excess over it. With the mismatch factor $M_s = (1-|\Gamma_s|^2)/|1-\Gamma_s\Gamma_{\rm rec}|^2$ and powers $P_{\rm ant} = g(M_{\rm ant}T_{\rm ant} + T_r)$, $P_{\rm load} = g(M_{\rm load}T_{\rm load} + T_r)$, $P_{\rm NS} - P_{\rm load} = gT_{\rm NS}$,

$$T_{\rm ant} = \frac{T^* - (1 - M_{\rm load})\,T_{\rm load}}{M_{\rm ant}}. \tag{2}$$

This is the minimal tier of M003 §7.1. It assumes equal gain on every switch path, no receiver noise waves, and that $T_{\rm NS}$ is the excess the receiver takes. Both steps are linear in the measured power, so a calibration is a pair of spectra, $T = s\,P + o$ (`eigsep_cal.dicke.tstar_coefficients`, `receiver_s11_coefficients`), that can be applied to any integration.

4 Inputs

4.1 Calibration visits

From 07-17 04:10 UTC the switch visits RFNON and RFAMB about every 13 minutes. Each visit becomes one spectrum: an integration-time-weighted mean over its samples. Samples are masked by the RFI bits of the

campaign flag product `flags@v3` (memo 006; released unchanged from `v3-beta.3`), which flags calibration rows against their own backgrounds, and, directly, when wrapped (int32 overflow) or in a dropped integration, which the flag product does not cover as RFI. At most 29 % of a visit's samples are masked. Two clock-aligned channels (125.0 and 187.5 MHz) are fully flagged in a few visits, and the calibration is NaN there for the integrations those visits bracket. 67 load and 65 noise-source visits are kept.

Receiver regimes. The calibration falls into two regimes, rx-A (to 07-17 19:43 UTC) and rx-B (from 07-18 01:25 UTC), separated by a 5.7 h gap with no usable calibration. Across the gap the noise source's injection changed abruptly while the load power did not (`marjum-2026-07/INDEX.md`); the names are historical, and no change in the receiver itself is established. Each visit carries the regime of its window from `curation/cal_windows.jsonl`, matched by file, because the window bounds are file close times. Calibration is never carried across the boundary.

Table 2 summarizes the 16/17 July night (21:30–07:55 MDT), the data set behind the spectra and comparisons below.

	16/17 July night
box-air integrations (input 4)	69849
correlator files	292
integrations timed from the filename (stale clock)	0
dropped (all-zero) integrations	0
integrations with an int32 wrap	289
channels with an int32 wrap	16
integrations without a load-thermistor reading	190
visits: antenna / noise source / load	54/48/49
longest time from a sky visit to the nearest load visit (min)	26.5
longest time from a sky visit to the nearest noise-source visit (min)	25.6
load temperature range (K)	298.46–301.53

Table 2: The 16/17 July night, box-air: what the calibration had to work with.

4.2 Contaminated visits

Three visits disagree with their neighbours in 278–296 channels by more than 10 %; no other rx-A visit exceeds 4 such channels, and rx-B's drifting edge visits reach 48. They are dropped when more than 100 channels disagree. Figure 2 shows what is in them: a comb of narrow lines at integer MHz, the odd-MHz teeth strongest. Fourier transformed across channels after removing a running median, each visit has comb power 6.6×10^5–8.0×10^5 times the noise at 1.000 MHz (4.096 channels) and at most 527 times at 1.953125 MHz (8 channels, the transmitter grid); their clean neighbouring visits show at most 2 times at either. This is the signature of box-air's 1.000 MHz comb EMI of 07-16 (memo 001), and Christian's notebook calls the 08:53 visit "the 1 MHz comb burst". Per integration it occupies the first minute or two of each visit and is absent on box-gnd (memo 006, § "The self-EMI in calibration visits on 07-17", which gives it per integration and masks it as `boxair-emi-1mhz`). It is not the transmitter, whose 8-channel comb first appears at 15:36:22 UTC. The identification rests on the spacing and on which antenna sees it, not on a physical test. All three visits also share a calibration window with an S11 sweep; whether that is related, and what switches the comb EMI on at the start of a calibration visit, are open (memo 006).

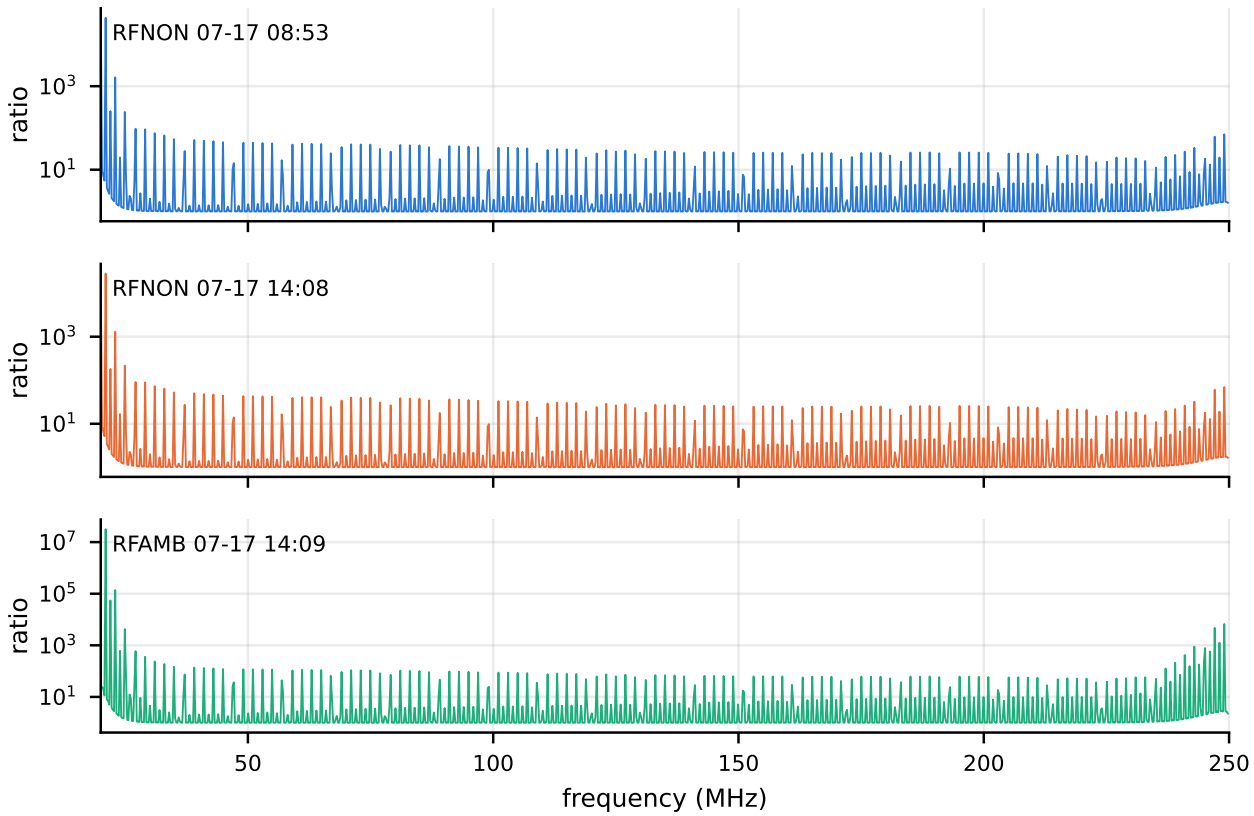


Figure 2: The three dropped calibration visits (raw integrations, weighted mean) divided by the mean of the two clean visits on either side (the visit spectra stored in tcal v0004).

4.3 S11 at plane P

Every field reflection is calibrated to P in two stages: a lab characterization of the internal standards and the switch paths, done once, and a per-sweep field calibration that uses it. Both use the one-port error model of Monsalve et al. (2016), in which a two-port network with S-parameters $(S_{11}, S_{12}S_{21}, S_{22})$ maps a reflection Γ at its port 2 to

$$\Gamma' = S_{11} + \frac{S_{12}S_{21}\,\Gamma}{1 - S_{22}\Gamma} \tag{3}$$

at its port 1. Embedding applies eq. 3; de-embedding inverts it. The network is solved from three standards of known reflection (open, short, load) measured through it. The known standards in the lab are a Copper Mountain Technologies S911T kit, whose open and short are modelled with their offset delay and loss (Monsalve et al. 2016, eqs. 18–21). The step-by-step procedure, with its code, is in `sparam_characterization.tex` in this directory and the notebook `EIGSEP.github.io/notebooks/characterizing_system_2026.ipynb` (commit `db669ac`); Appendix B shows its intermediate results.

Lab (2026-09-12). Both lab steps use the field switch matrix and the S911T standards; the raw captures are in `data/s11/cal_materials/lab/` (their file names, dated 2026-06-21 and 2026-06-29, carry the recording computer's wrong clock). First, S911T on the free end of the VNA cable solves the VNA-and-cable network; removing it from the matrix's internal open, short and load, measured through the cable, puts them at @VNA (Figure 13). Second, S911T at each source's port, and at the LNA port, is measured through each `VNA*` path from @VNA and through each `RF*` path from P; with the VNA-and-cable network removed, each set of three solves that path's S-parameters (Figure 3). In the band the paths lose 0.5–1.4 dB and reflect below −18 dB at both ports. Rederived from the raw captures, the internal standards reproduce `eigsep_cal`'s packaged `calibrated_internal_osl2026.npz` to 9×10^{-16}, and the seven paths

`data/s11/cal_materials/switch_sparams.npz` (packaged in `eigsep_cal` since 96d4e9e) to 1×10^{-15}. The noise diode's paths are not among them (§2).

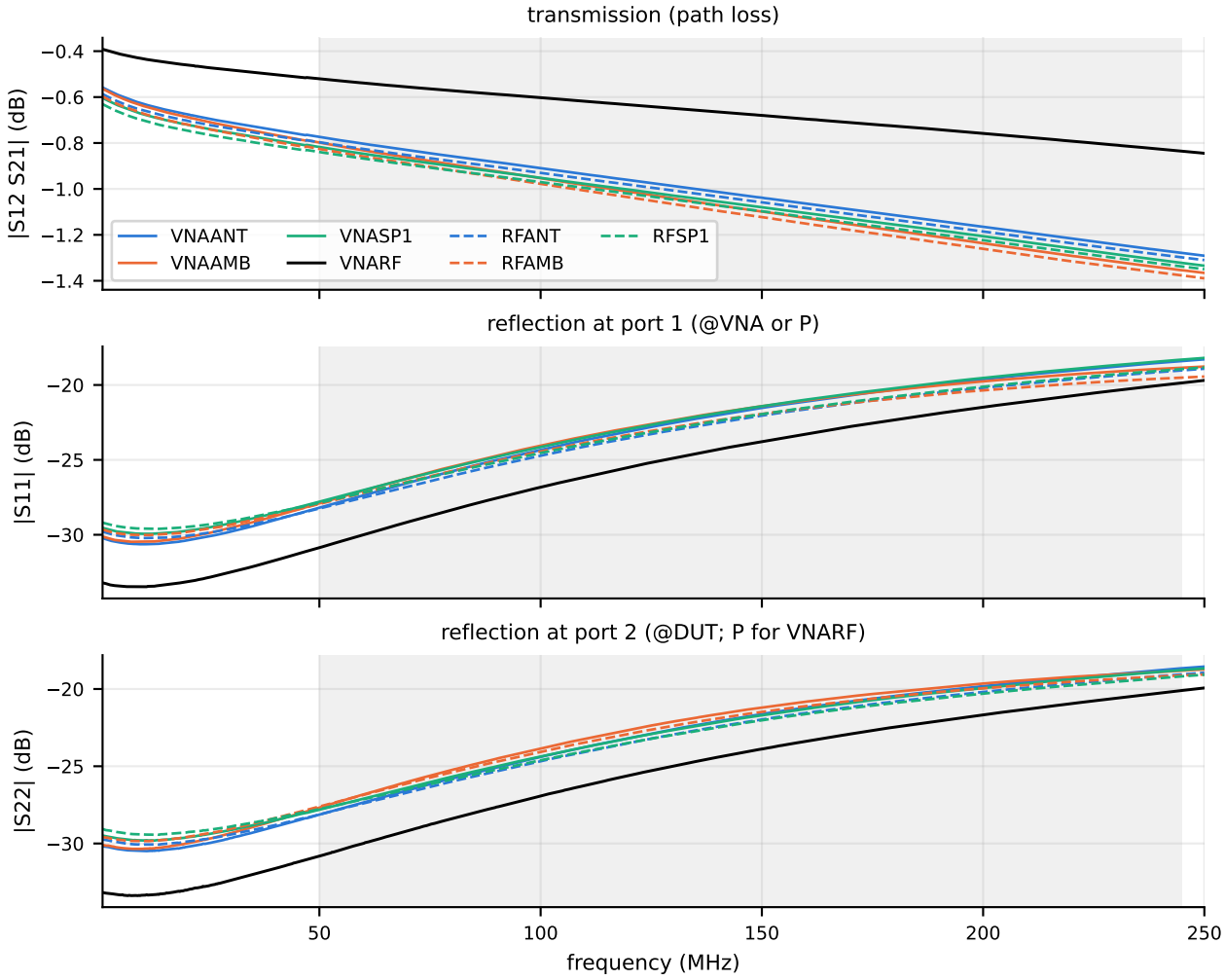


Figure 3: The seven characterized switch paths (`switch_sparams.npz`): transmission $|S_{12}S_{21}|$, the path's round-trip loss (top), and the reflections at each end (middle, bottom). Colour is the source port (antenna, ambient load, SP1); solid paths join @VNA to a source port (`VNA*`), dashed paths join P to it (`RF*`), and `VNARF` (black) joins @VNA to P. The calibration band is shaded. Each path was solved in the lab on 2026-09-12 from S911T standards measured through it.

Field: each sweep. 40 raw VNA sweeps (`data/s11/`) are calibrated with `eigsep_cal.S11`. Each sweep records the internal standards with its sources. The VNA network at the time of the sweep is solved from them against their lab characterization, and de-embedded from every source in the sweep, which puts each at @VNA. A sweep whose own internal standards are zero-filled borrows the nearest valid set. From @VNA the chain depends on the source:

- **Sources with a port** (`amb`, `ant`, `sp1_open`, `sp1_short`): de-embed the source's `VNA*` path, which gives its reflection at @DUT, then embed its `RF*` path, which gives it at P.
- **The receiver** (`rec`): de-embed `VNARF`, which ends at P already. There is nothing to embed.
- **The noise diode, on and off** (`noise`, `load`): neither path can be characterized, so these stop at @VNA. Nothing in the calibration uses them.

Figure 14 follows one antenna and one receiver sweep through these steps. This agrees with Christian's independent chain to 1×10^{-15} in Γ at P.

4.4 Load temperature and noise-source excess

T_{load} is the `tempctrl_load` thermistor reading nearest each integration (logged whether or not the switch cycles; 300.0 K on the night). The noise diode (Noisecom NC302L) is specified only as ENR 30–35 dB, with no flatness given: an excess of 290–917 K behind its 30 dB pad. `tcal` up to v0003 took the top of that range. From v0004, T_{NS} is one spectrum, applied to every integration, built from the bench measurement of §7 (`noise_source` v0001):

$$T_{\mathrm{NS}} = M_{\mathrm{NS}} \left(T_{\mathrm{load}} + T_{\mathrm{ex}}\right) - M_{\mathrm{load}} T_{\mathrm{load}}, \tag{4}$$

the power the diode path adds over the load's as the field receiver takes it (`eigsep_cal.noise_source.tns_as_received`). T_{ex} is the diode's excess at P, the smooth fit of the bench result; M_{NS} uses the lab Γ_{NS} at P; M_{load} and M_{NS} use the field Γ_{rec} and Γ_{amb} averaged over the campaign's 16 sweeps of each; and T_{load} is the campaign's mean load thermistor, 302.5 K, which takes the diode path's temperature when off to be the load's. Over 50–200 MHz T_{NS} = 489–600 K. Built from each receiver sweep's own reflections instead of the means it would move by at most 0.16 % (0.02 % rms), so one spectrum serves every integration. This also puts the noise source's own port mismatch into the calibration, which v0003 took as 1.

5 The tcal product

5.1 How each integration is calibrated

For each integration, P_{load} and P_{NS} are interpolated linearly in time between the visits that bracket it, within one receiver regime and only across a gap of at most 90 min. Each Γ is interpolated between the bracketing sweeps of its epoch (gap at most 3 h), or held from the nearest sweep within 2 h; sweeps are never used across the 07-17 19:11 → 07-18 01:17 gap, which spans the regime boundary. T_{load} is the nearest thermistor reading within 60 s. Anything outside a limit is NaN. The calibrated view is NaN on rows that are not `RFANT`; `Bundle.t_star` gives T^* for any state.

5.2 Why these limits

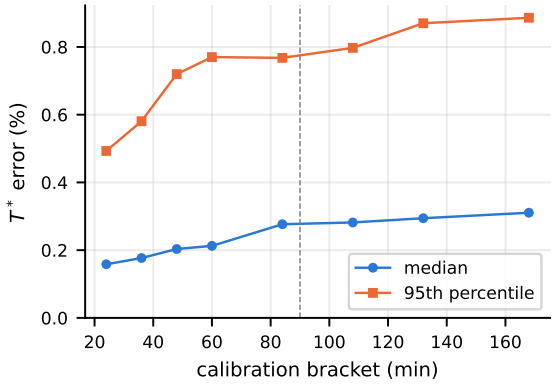
Leave-one-out tests (`validation.json`) predict each interior visit from its neighbours. For the gain proxy $P_{\mathrm{NS}} - P_{\mathrm{load}}$ the median error is 0.21 % linear and 0.33 % for the nearer neighbour; linear wins in 71 % of cases. Widening the bracket (Figure 4) costs little: the median error levels off at 0.28–0.31 % beyond about 90 minutes. Holding an S11 sweep for 1 or 2 h moves the calibrated temperature by 0.03 and 0.04 % (95th percentile at most 0.13 %).

The 177-minute gap in rx-A (07-17 16:22 → 19:19) is not bridged. The gain proxy changes by 8.4 % across it, against a median of 2.4 % and a maximum of 5.3 % over the same span elsewhere; the antenna was raised to 91 m inside it. The antenna S11 changes by 2.1×10^{-3} across it, inside its usual range (95th percentile 3.5×10^{-3}).

5.3 Coverage, drift and validation

Figure 5 (top) shows when the calibration of the product is well defined according to the visit interpolation and calibration rules (§5.1). The lower panel shows why it interpolates between visits and does not bridge gaps: the noise source's injection, relative to the load power, which is immune to receiver gain, rises 16.3 % peak to peak over the 12.2 h before the 177-minute rx-A gap (correlation with time +0.90), while the load power holds to 0.85 % rms. Across the gap it steps up further. Interpolating between visits follows this drift; pairing each integration with its nearest visit does not. This reproduces the noise-source drift of legacy MEMO-004 independently.

Through `load_bundle`, T^* on load rows returns the thermistor temperature to a median of 0.003 K (95th percentile 0.7 K) and on noise-source rows $T_{\mathrm{load}} + T_{\mathrm{NS}}$ to 0.16 K (4.9 K). The worst residuals are the contaminated rows themselves.



bracket (min)	median (%)	95th pct (%)	max (%)
24.0	0.16	0.49	0.8
36.0	0.18	0.58	1.1
48.0	0.20	0.72	1.3
60.0	0.21	0.77	1.7
84.0	0.28	0.77	1.8
108.0	0.28	0.80	1.7
132.0	0.29	0.87	1.9
168.0	0.31	0.89	1.9

Figure 4: Error in T^* from interpolating the calibration across a wider bracket, on rx-A (50–200 MHz, on the night's mean sky). The dashed line is the 90-minute limit.

Christian's calibration of the night (2026-09-25) is independent of this one in code but takes $T_{\mathrm{NS}} = 917$ K. `tcal` v0003, identical to v0004 but for that value, reproduces it to a median ratio of 1.0003 per sky visit over 50–200 MHz. With the bench T_{NS} the ratio is 0.7698 (5th–95th percentile 0.7427–0.8300), the change of scale and nothing else. Both sides take per-visit medians there, because Christian's chain does.

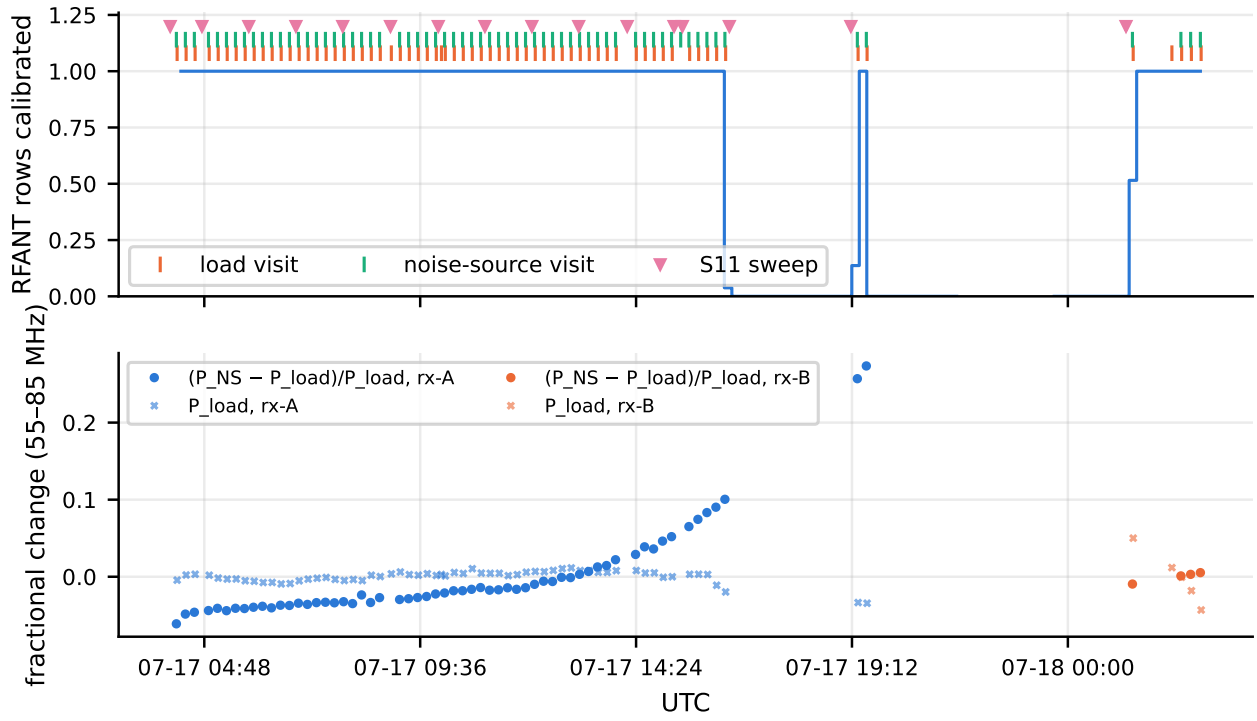


Figure 5: Top: fraction of box-air antenna integrations calibrated, per 10 minutes, with the calibration visits and S11 sweeps. Bottom: noise-source excess over load power and load power, 55–85 MHz, relative to their means, for each receiver regime (rx-A, rx-B).

6 The spectrum, and where the scale fails

Figure 6 shows the night's calibrated spectrum: weighted means over antenna integrations, with the `flags@v3` RFI bits and wrapped samples as zero weight. The RFI bits mask 8.8 % of antenna samples on the night and 68 % in the FM band. They do not cover int32 wraps: 227 wrapped samples, each near -2×10^9 counts, would otherwise put deep false dips in the aircraft band. A channel with less than half its integration time left after masking is dropped (74 channels, most of the FM band), because the RFI bits flag mostly positive excursions and the remainder is biased low.

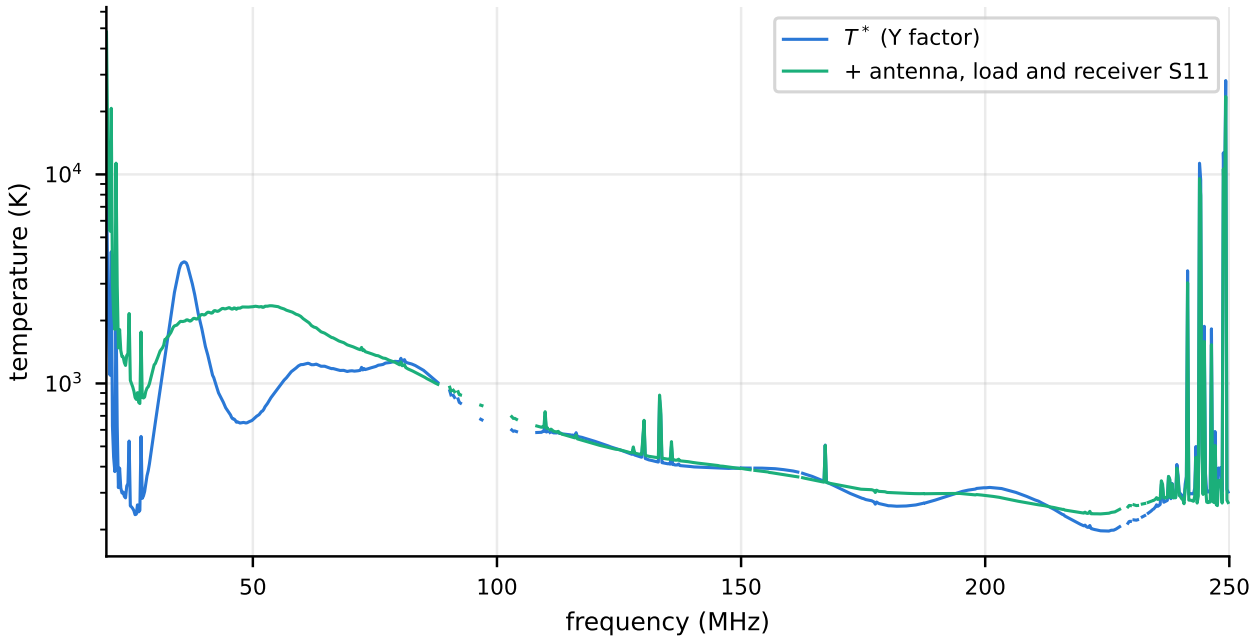


Figure 6: The 16/17 July night through tcal v0004: T^* and the fully corrected temperature, weighted means over antenna integrations.

Below about 30 MHz the scale has failed. The receiver passes almost nothing there: the load power is zero counts at 5–15 MHz (Table 3). When the antenna and load powers round to the same count, T^* returns T_{load}; where the noise-source and load powers are equal it divides by zero. The values near 300 K are the formula, not the sky.

From 30 to about 60 MHz the result rests on the reflection corrections. The antenna reflects most of its power at P ($|\Gamma_{\mathrm{ant}}| \approx 0.7$–$0.8$), $|\Gamma_{\mathrm{rec}}|$ peaks near 0.7, so the reflection corrections are large there (M_{ant} in Table 3). The bump in T^* near 36 MHz is removed by the receiver term ($M_{\mathrm{ant}} > 1$ there). The correction treats every reduction of $|\Gamma|$ at P as mismatch; but the balun, coax and switch path between the antenna and P absorb power and re-emit at their physical temperature, which the S11 cannot see. That loss is largest where the antenna is worst matched. Treat the band below about 60 MHz as uncalibrated until the balun's loss is bounded; its S-parameters cannot be measured directly (§8).

Against a sky model. Figure 7 compares the calibrated spectrum with GSM2016 seen through the HFSS beam normalized to directivity, pointing at zenith as it did that night, with 300 K ground filling the lower hemisphere (the bowtie radiates 0.41–0.63 of its power upward). The simulation rises monotonically toward low frequency; the data flatten below 60 MHz and fall below it there (data/simulation 0.68 at 51 MHz, 0.89 at 59), meet it near 70 MHz (0.99), and sit just above it from 78 MHz: 1.10 at 78, 1.06 at 129 and 1.08 at 148 MHz. With the 917 K of tcal v0003 they sat 1.29–1.66 times high at 78–150 MHz (§7). The simulation is

MHz	P_{load} (counts)	$\lvert\Gamma_{\mathrm{ant}}\rvert$	M_{ant}	T^{*} (K)	T (K)
4.9	9.85	0.83	0.50	751	1542
10.0	2.38	0.81	0.29	743	2604
20.0	2.45	0.74	0.20	9754	47953
24.9	701	0.72	0.25	259	1035
30.0	2.8e+04	0.71	0.52	652	1246
34.9	9.62e+04	0.72	1.78	3515	1965
40.0	1.98e+05	0.74	0.74	1588	2126
50.0	4.42e+05	0.74	0.28	673	2329
60.1	6.04e+05	0.62	0.59	1230	2063

Table 3: The low band on the night: mean load power, antenna reflection and mismatch at P (night-mean sweep), and the weighted-mean temperatures.

at the antenna terminals and the data at P, so the comparison cannot separate what remains: an error in T_{NS} scales the data about T_0 (§7), while balun and coax loss and noise waves change its shape.

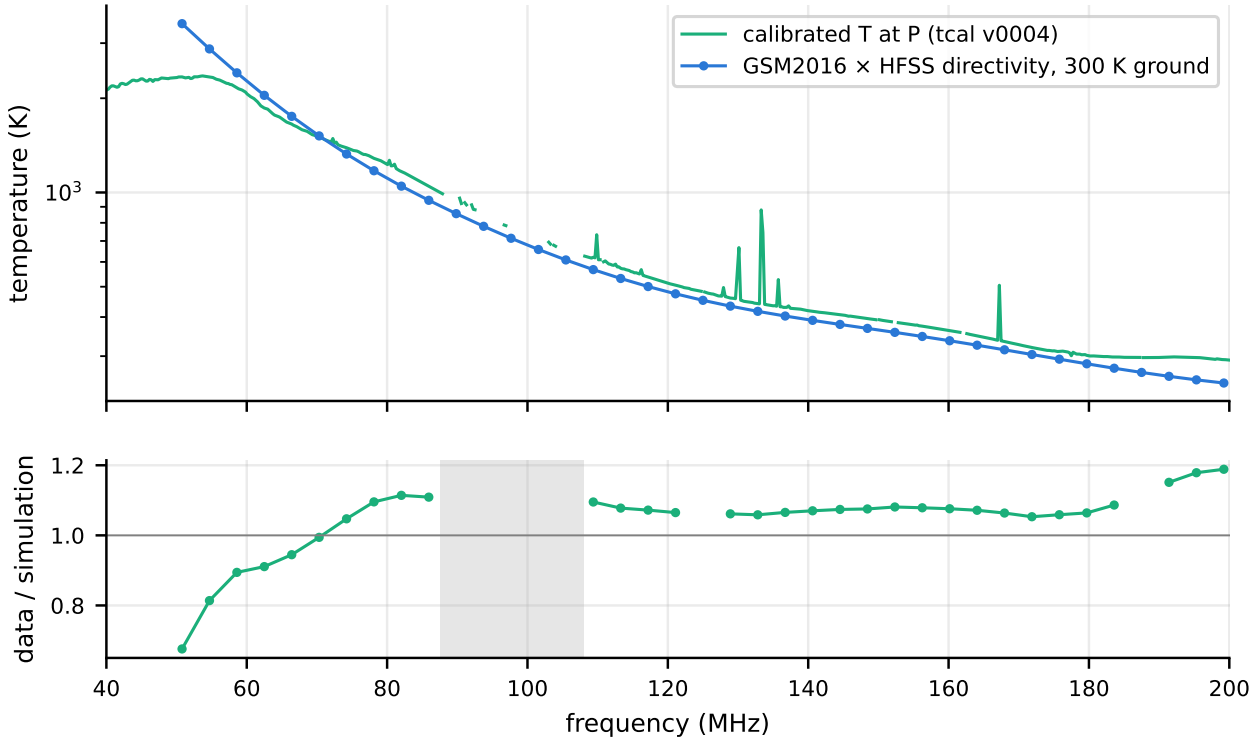


Figure 7: Calibrated temperature on the night against GSM2016 × HFSS directivity, weighted by the calibrated integrations in each 15-minute bin. The FM band is left out of the ratio.

The HFSS "efficiency" is port mismatch. The HFSS gain integrates to well under 4π at low frequency. Dividing by the mismatch of HFSS's own input impedance against a 75 Ω port leaves a radiation efficiency near 1 from 51 to 100 MHz (Table 4); against 50 Ω it is not flat. So the HFSS port was 75 Ω, the simulated radiator has negligible ohmic loss, and the HFSS gain must not be applied on top of the measured mismatch correction.

MHz	Z_{HFSS} (Ω)	$\int G\,d\Omega/4\pi$	$1-\lvert\Gamma_{75}\rvert^2$	e_{rad} (75 Ω)	e_{rad} (50 Ω)
51	$18.5-61.5j$	0.453	0.444	1.02	1.04
59	$21.3-42.5j$	0.562	0.576	0.98	0.91
70	$26.6-20.8j$	0.702	0.742	0.95	0.83
78	$31.9-7.1j$	0.791	0.834	0.95	0.84
86	$39.0+5.3j$	0.865	0.898	0.96	0.88
100	$56.9+23.0j$	0.959	0.952	1.01	1.01

Table 4: HFSS realized gain split into port mismatch and radiation efficiency (`eigsep_sim` bowtie v000 impedance; the beam integral is `bowtie_beam.npz`). Z_{HFSS}: the simulated antenna's input impedance at its port. $\int G\,d\Omega/4\pi$: the HFSS realized gain averaged over the sphere, the fraction of the power offered at the port that is radiated. $1-\lvert\Gamma_{75}\rvert^2$: the fraction of that power accepted by the antenna, with Γ_{75} its reflection against a 75 Ω port. e_{rad}: radiation efficiency, $\int G\,d\Omega/4\pi$ divided by $1-\lvert\Gamma\rvert^2$ against a 75 Ω or a 50 Ω port.

7 Bench runs: the noise-source temperature

7.1 The 2026-10-08 run

The field cannot measure T_{NS}: it has one load at one temperature, and the noise diode sits on the switch board with no connector. On the bench, a hot load of known temperature provides the second point. The interim run of 2026-10-08 (PDT; UTC 2026-10-09 01:01–04:41, 3.7 h) used the field switch board, with a heated 50 Ω load on the `RFAMB` port, an ambient reference load on the `RFANT` port, and a lab LNA chain in place of the field receiver. The switch cycled reference, hot load, noise source on and off, 10 s each, while the hot load stepped through three heater plateaus and then cooled. The correlator recorded 12,296 integrations of 1.074 s in 72 files, with VNA sweeps of the loads and the receiver during the run. The wiring and its departures from the full measurement are in the run record, `data-analysis/notebooks/charlie/bench_runs/2026-10-08_interim/`. The run answered five questions:

1. *Do the thermistors follow the hot load?* Within this test, yes (Figure 8): cycles in which the hot-load thermistor changes by more than 1 K/min give slopes $0.3\,\sigma$ (heating) and $-0.3\,\sigma$ (cooling) off the steady cycles' (mean z). Only the 153 steady cycles are fitted all the same (hot load 22–87 °C); the 15 heating and 19 cooling cycles are not.
2. *Is the switch sequence clean, and which thermistor is the diode's?* Figure 9: each state holds a steady power for its dwell, and the switch board's `temp_therm2` (GP28) warms during each `RFNON` dwell, so it is the diode's thermistor on this board, as on the field board. Integrations that straddle a switch change are recorded as `UNKNOWN` and never used.
3. *Does the VNA disturb the radiometer?* Yes: in the seconds before many sweeps, every state's power rises well above its dwell level. Integrations from 20 s before to 5 s after each of the 10 sweeps are excluded.
4. *What is the noise source's temperature at P?* 491–589 K of excess across the band (Table 5, Figure 11).
5. *Does it transfer to the field receiver?* Yes, once the noise source's own reflection is removed: the result is then a property of the switch board, not of the receiver it was measured with (below).

7.2 Method

Each dwell is an integration-time-weighted mean, with samples flagged by `eigsep_data.flagging`'s detectors (int32 wraps, transients) and the VNA guard above. For each switch cycle the switched ratio $Q = (P_{\mathrm{hot}} - P_{\mathrm{ref}})/(P_{\mathrm{NON}} - P_{\mathrm{ref}})$ cancels the gain. With the model of §3, $P_s = g(M_s T_s + T_r)$ for both loads and $P_{\mathrm{NON}} = g(M_{\mathrm{NS}} T_{\mathrm{NON}} + T_r)$ for the diode path when on,

$$M_{\mathrm{hot}} T_{\mathrm{hot}} - (1-Q)\, M_{\mathrm{ref}} T_{\mathrm{ref}} = T_{\mathrm{NON}}\, Q\, M_{\mathrm{NS}}, \tag{5}$$

so T_{NON}, the diode path's temperature at P when on, is the slope of a straight line through the steady cycles, fitted per channel (`eigsep_cal.noise_source.fit_source_temperature`). T_{hot} and T_{ref} are the two loads' own thermistors, on pins GP26 and GP27 (GP27 is logged as `tempctrl_lna1`, although the run's config labels

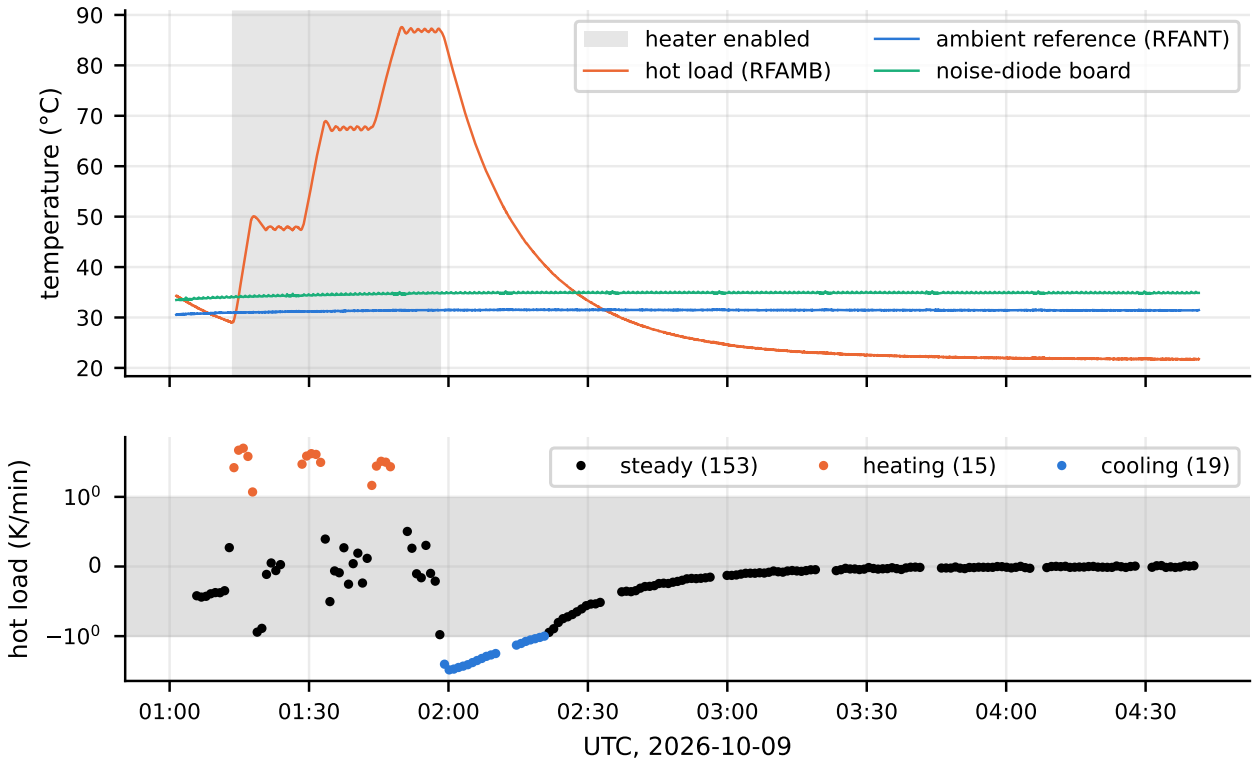


Figure 8: The 2026-10-08 bench run. Top: the hot load's, the ambient reference's and the noise-diode board's thermistors, with the heater's enabled periods shaded. Bottom: the hot-load thermistor's rate of change in each switch cycle (symmetric log scale), by how the cycle is classified; the grey band is the steady limit, and only cycles inside it are fitted. The heating ramps after each new set point fall outside it.

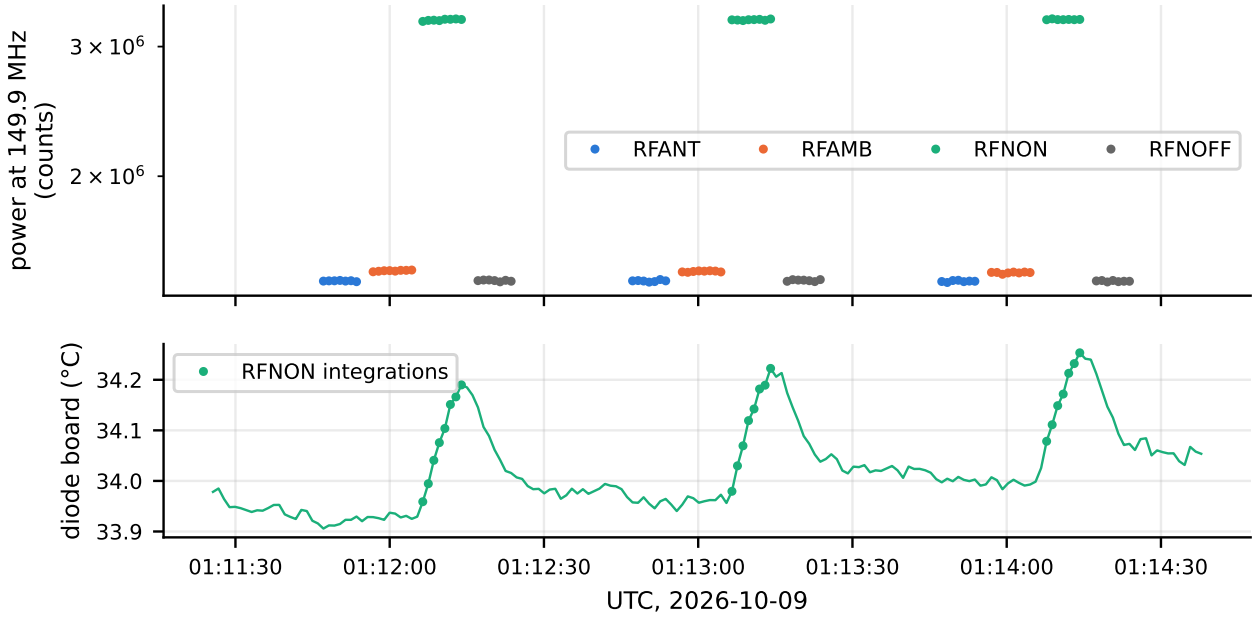


Figure 9: Three switch cycles, ten minutes into the bench run. Top: raw power of each integration at 149.9 MHz, by switch state. Bottom: the switch board's `temp_therm2`, which warms in every `RFNON` dwell, identifying it as the noise diode's thermistor.

it `tempctrl_lna2`, which is GP28, LNA 3's: before the heater first comes on, the two loads' power difference matches GP27's reading and not GP28's). Taking T_{ref} from GP28 instead, as the product's first build did, gives an excess ∼1 % lower and heating cycles that disagree with the steady ones. The loads' reflections at P come from the run's own sweeps through the S11 chain of §4.3; both de-embed to their own ports with $|\Gamma| \leq 0.006$, so the packaged switch-path characterization fits this board. Γ_{NS} at P was measured in the lab on 2026-10-15 with the VNA on the board's LNA port and the switch on `RFNON` (Figure 10); with the diode off (`RFNOFF`) it agrees to 0.0016, as it should behind a 30 dB pad. Rederived from the raw capture (`cal_materials/lab/`) it reproduces the `rfn_sparameters_10_15.npz` the measurement reads to 6×10^{-16}. $|\Gamma_{\mathrm{NS}}|$ is 0.028–0.094 in the band, and M_{NS} with the bench receiver 0.97–1.08. The excess is $T_{\mathrm{NON}} - T_{\mathrm{off}}$, with $T_{\mathrm{off}} = 308.0\,\mathrm{K}$ (35 °C) from the diode board's thermistor during the noise-off dwells.

A channel is masked if it is outside 50–245 MHz (225), carries a steady line in any state (85), has fit residuals more than three times the in-band median in radiometer units (217), or has a slope error over 5 % (46); 575 channels remain. Q scatters between cycles 1.2 times its radiometer noise (median χ^2 1.4), which the slope's error does not include. The measurement is `eigsep_data.noise_source_bench.measure`, run by `eigsep-build-noise-source` with the choices in the product's `config.json`; `make_products.py` runs it again on the checksummed bench files and reproduces the product exactly.

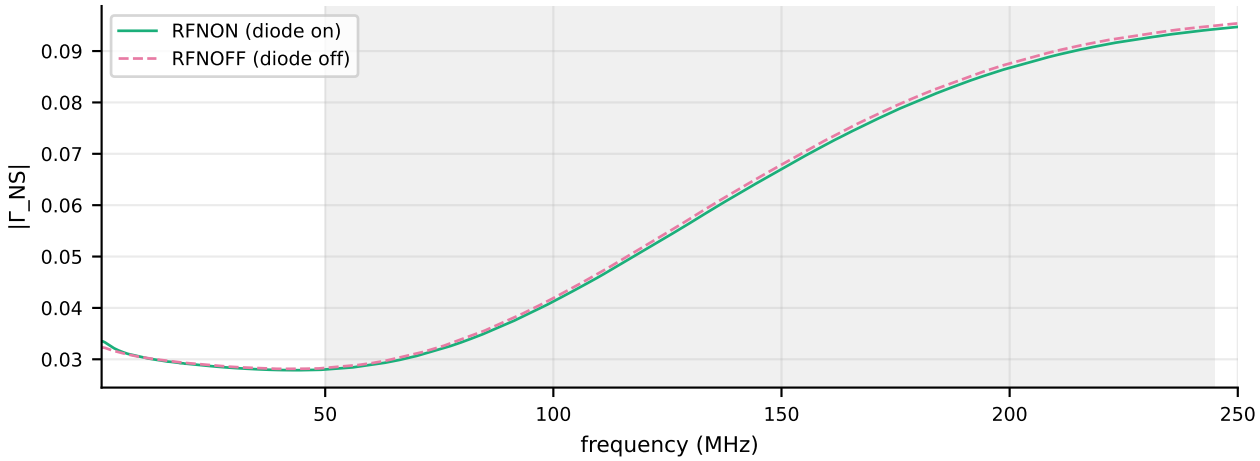


Figure 10: The noise-source path's reflection at P, measured in the lab on 2026-10-15 with the VNA (its cable calibrated with S911T standards at the free end) on the field switch board's LNA port, with the switch on `RFNON` (solid) and `RFNOFF` (dashed). The two agree to within 0.0016, as they should with the diode behind a 30 dB pad, so one Γ_{NS} serves both states. The band is shaded.

7.3 Result

Figure 11 follows T_{NS} through the analysis. Fitted against the thermistor difference alone, the slope of Q gives 512 K (weighted mean, 50–245 MHz). With the loads' reflections it gives 522 K; this still absorbs the noise source's own mismatch with the bench receiver, so it would not transfer. Equation 5 removes that, and the excess at P is 510 K, falling with frequency from 589 K below the FM band to 491 K above 200 MHz (Table 5). Were it the excess behind the pad, that is ENR 32.3–33.1 dB: inside the specified 30–35 dB and below its top. A cubic in frequency, weighted by the channel errors, carries it to every channel (residual rms 4.8 K, worst 15 K, at 216 MHz) and is what `tcal` uses, through equation 4, because the field and bench channel masks differ. It is published as `marjum-2026-07/derived/noise_source/v0001/`, read with `eigsep_data.products.noise_source`.

MHz	channels	T_{NON} (K)	excess (K)	per channel (K)	ENR (dB)
50–88	78	897	589	561–623	33.1
108–150	166	831	523	509–537	32.6
150–200	175	804	496	485–510	32.3
200–245	134	799	491	476–504	32.3

Table 5: The noise source at P from the bench run (`noise_source` v0001): inverse-variance weighted means over the unmasked channels of each band, the per-channel range of the excess, and the ENR it would be behind a 30 dB pad.

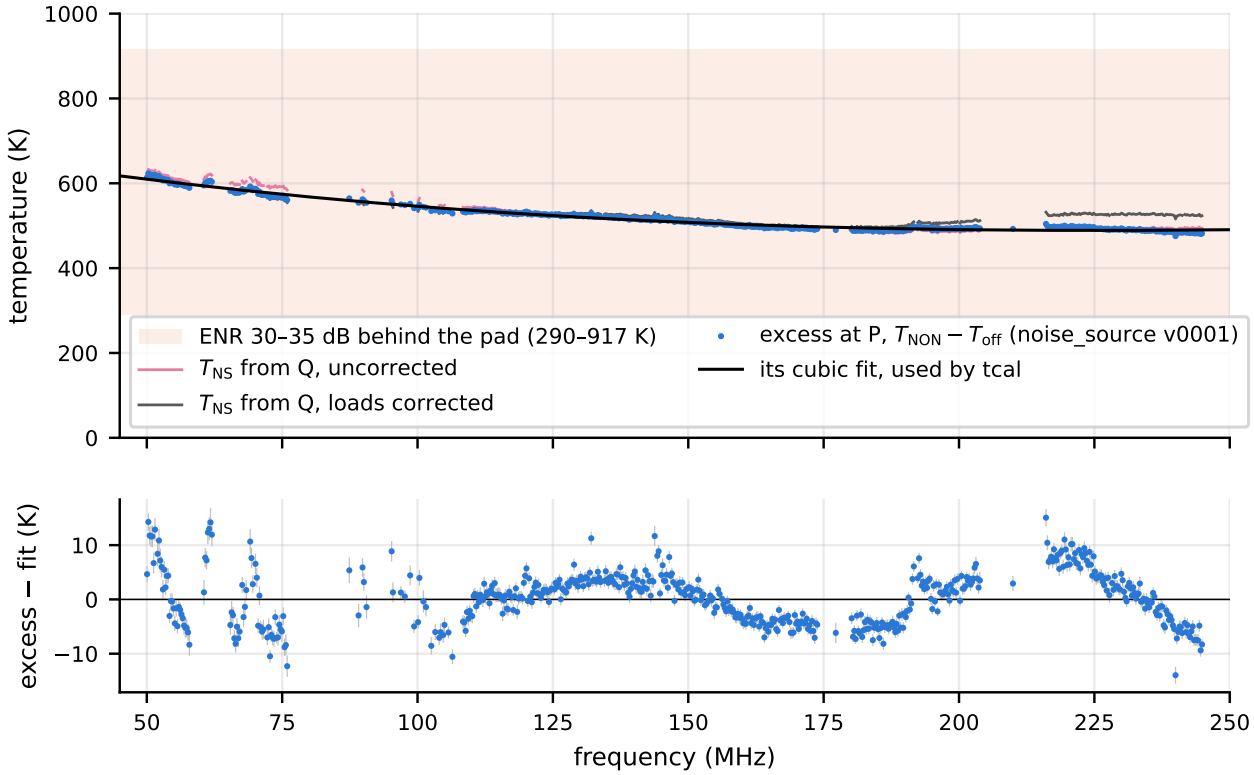


Figure 11: The noise source's temperature from the steady bench cycles, masked channels blank. Pink and grey: T_{NS} from the slope of Q, against the thermistor difference (pink) and with the loads' reflections at P (grey); both absorb the noise source's mismatch with the bench receiver. Blue: the excess at P with that mismatch removed (equation 5), `noise_source` v0001; black: its cubic fit, which `tcal` v0004 uses. Shaded: the excess an ENR of 30–35 dB gives behind the 30 dB pad. Bottom: the excess minus the fit, with each channel's error.

7.4 In the field

Figure 12 shows what the measured T_{NS} does to the night of §6. Each channel's calibrated temperature is linear in T_{NS} (`eigsep_cal.noise_source.rescale_tns`),

$$T(T_{\mathrm{NS}}) = T_0 + \frac{T_{\mathrm{NS}}}{T'_{\mathrm{NS}}}\left(T(T'_{\mathrm{NS}}) - T_0\right), \qquad T_0 = \frac{M_{\mathrm{load}} T_{\mathrm{load}}}{M_{\mathrm{ant}}}, \tag{6}$$

so changing T_{NS} moves every channel along a line through its own T_0, the calibration at $T_{\mathrm{NS}} = 0$. Whether a channel lies above or below T_0 is the sign of $P_{\mathrm{ant}} - P_{\mathrm{load}}$ and does not depend on T_{NS}. Where the sky is far above T_0, at low frequencies, the diode's specified range spans a wide range of temperatures; where the data sit near T_0 it pinches. In 90 channels between 172 and 194 MHz the data sit below T_0 ($M_{\mathrm{load}}/M_{\mathrm{ant}} > 1$ there), and a smaller T_{NS} raises them. Against GSM2016 × HFSS at 78–150 MHz the data sit 1.06–1.11 times the simulation

with the bench T_{NS}; the diode's specified range would allow 0.69–0.96 at 30 dB to 1.28–1.64 at 35 dB, and 917 K gave 1.29–1.66.

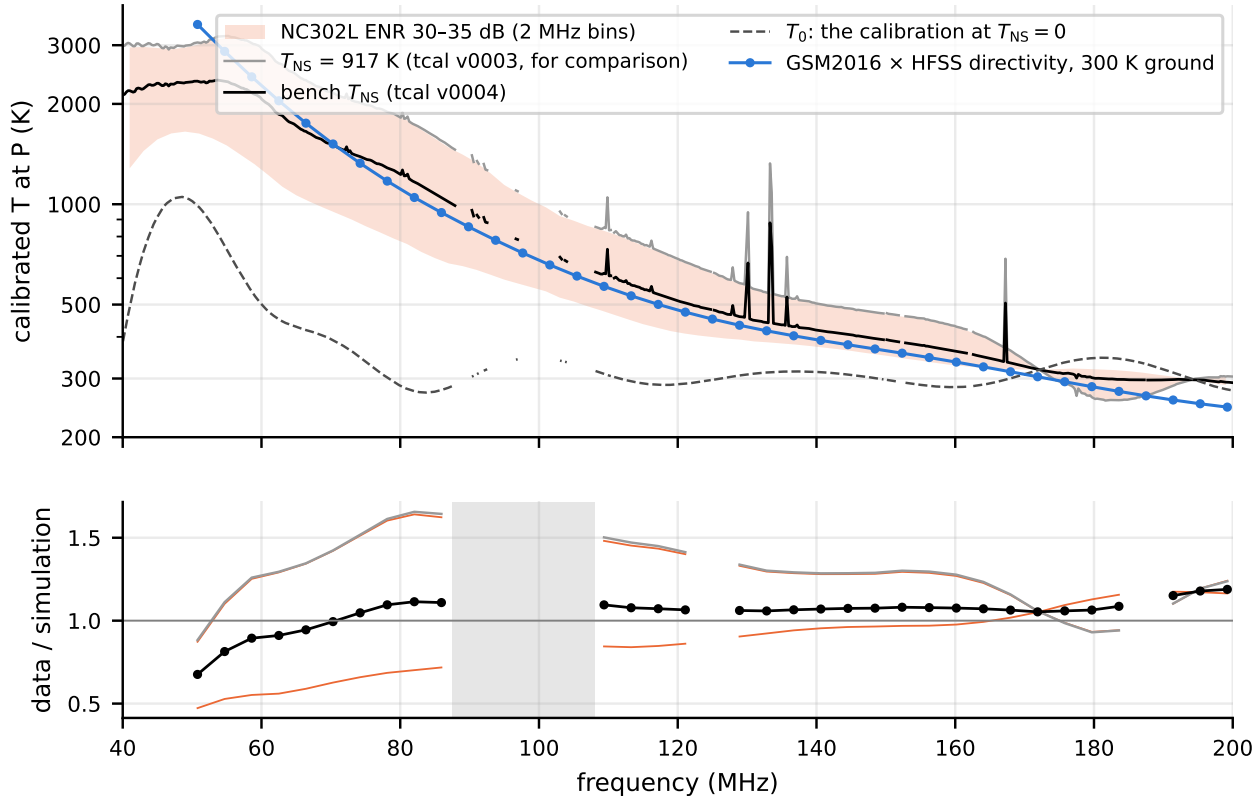


Figure 12: The 16/17 July night's calibrated temperature at P (weighted means over antenna integrations, as in Figure 6) for the noise-source temperatures in play. Black: the bench T_{NS} (tcal v0004). Grey: $T_{NS} = 917$ K, as tcal v0003 had it, for comparison. Shaded: ENR 30–35 dB converted to the field receiver with equation 4, as weighted means over the night's unmasked channels in 2 MHz bins; 9 channels with narrow residual lines, more than 5 % off the running median of their 15 neighbours, are left out of the band so it does not follow them. Dashed: T_0, the calibration at $T_{NS} = 0$, which every channel pivots about. Blue: GSM2016 × HFSS directivity with 300 K ground, as in Figure 7. Bottom: data over simulation; the FM band is left out.

7.5 What the bench result does not yet cover

- **One board temperature.** The diode board sat near 35 °C on the bench; the field load sat near 302.5 K. How the diode's excess and Γ_{NS} change with temperature is not measured, and in the field the injection drifts through the night (§5). Interpolating between calibration visits follows that drift; the absolute T_{NS} is the bench value.
- **No switch-path loss.** The `RFNON` path's loss sits inside the excess at P, which is what the calibration needs; but equation 4 takes the path to be at the load's temperature when off.
- **Thermistors** are read through 2-wire pull-ups measured on the bench, with ±0.2 °C parts.
- **The bench receiver** is not the field receiver. T_{NON} should not depend on it, but this is untested against a second receiver.

8 Open questions

- **The noise-source temperature.** Measured once, on the bench, at one diode board temperature (§7); it sets the scale of `tcal` v0004 and is still its largest uncertainty. The field injection drifts through the night (§5), and whether that is the diode's excess or its path's reflection changing with temperature is not known. Repeating the bench run at several board temperatures, with the field receiver in place of the lab LNA, would measure the first and test the transfer to the field receiver directly; bounding the `RFNON` path's loss would test equation 4's assumption that the path sits at the load's temperature. The bench measurement replaces the in-band estimate of legacy MEMO-004, which treated the ambient load and the diode-off path as two passive loads at their thermistor temperatures (the noise diode's being `rfswitch_therm.temp_therm2`, GP28), gave a value 2.4 times below 917 K, and failed a closure test by the same factor.
- **Temperature-dependent Γ_{NS}.** The noise-source path's switch S-parameters (`VNANON`, `RFNON`) cannot be measured, because the diode sits on the switch board with no connector (§2), but its reflection at P can, and that is all the calibration needs. It was measured once, for §7. Measurements of Γ_{NS} at P (VNA on the board's LNA port, switch on `RFNON`) against the board's temperature are under way. With $|\Gamma_{\mathrm{rec}}|$ up to 0.7, a small change in Γ_{NS} moves M_{NS}, and so T_{NS} (equation 4), with the board's temperature; the measurements say whether that accounts for part of the field drift, and give M_{NS} at the field board's temperature from its thermistor.
- **Receiver noise waves.** Omitted from `tcal`. With $|\Gamma_{\mathrm{ant}}|$ and $|\Gamma_{\mathrm{rec}}|$ both large at the lower end of the band they cannot be negligible there. A first tier-2 fit (M003 §7.1) to the 07-18 rx-B spare-port open/short blocks and the ambient load, with T_{NS} fixed, is in `data-analysis/scripts/marjum-2026-07/noisewave/` (README for the numbers). Smooth noise-wave parameters do not fit those data; allowing the noise waves a phase delay between P and the LNA predicts a held-out termination far better but leaves a residual well above the block-to-block repeatability. That residual can be absorbed either by a consistent error in the measured receiver reflection or by structure in T_{unc}, and the field data cannot tell which. A bench measurement of the receiver's reflection at P, through the same switch path, decides it; the study's README lists the measurements in order. Until then tier-2 temperatures do not replace tier 1.
- **Balun and coax.** Their available gain belongs in the comparison with any simulation. The balun board's S-parameters cannot be measured: its interface to the antenna is not a uniform, connectorized port, so no set of open, short and load standards can be defined there, and no two-port calibration can connect to it in a characterizable way either. Its loss has to be bounded by other means. The manufacturer's four-port S-parameter file (`.s4p`) for the balun chip is available; it describes the chip alone, not the board's traces or its interface to the antenna. The balun–switch coax did not survive; a spare of the same type and length could be measured.
- **RFI mask.** `flags@v3` (memo 006) removes most of the lines `flags@v2` left, but some aircraft-band and band-edge lines remain in the night's spectrum, and its RFI bits do not cover int32 wraps; mask negative samples as well.
- **Coverage.** rx-A 16:22–19:19 and the regime gap are NaN. Bridging them needs a model of the drift across the antenna raise, not a wider limit.

Revision history

- 2026-10-23: the bench measurement takes its reference temperature from the reference load's own thermistor (it had been LNA 3's); the noise source's excess rises by ∼1 %, and the heating cycles now agree with the steady ones, so the earlier conclusion that the thermistor lags the heater is withdrawn. The S11 calibration is shown in figures (§4.3, Appendix B) and checked against its raw lab captures, which were taken on 2026-09-12, not on the dates their file names carry.
- 2026-10-20: the noise-source temperature is measured on the bench (§7, `noise_source` v0001) and the memo moves to `tcal` v0004, which uses it in place of the assumed 917 K. The scale drops by a factor of 0.53–0.65 in $T^* - T_{\mathrm{load}}$, and the data/GSM × HFSS excess at 78–150 MHz largely goes; Christian's calibration is now a check of the code at 917 K, not of the scale.

- 2026-10-05: built on tcal v0003 and `flags@v3`; shows current results only.
- 2026-10-03: the contaminated calibration visits identified as box-air's 1.000 MHz comb EMI; the noise-wave open question points to the first tier-2 study.
- 2026-10-01: first version. Restates legacy MEMO-004 (noise-source drift, reproduced independently in §5; the in-band T_{NS} route in §8) and the calibration method of legacy MEMO-003. MEMO-003's receiver-temperature stability and uncertainty budget are not yet restated here, so that file stays until they are.

A Reproducing this memo

Run `python make_products.py` in this directory, with `EIGSEP_CAMPAIGN_ROOT` pointing at a campaign tree that holds the raw data, then `pdflatex memo-004.tex` twice. Tracked products are read from this checkout. The script checks the raw S11 files against `data/s11/manifest.csv` and the bench run's files (`data/bench/2026-10-08_interim/`) against the `noise_source` manifest, records a digest of the correlator files it reads, and writes every figure in `figures/` and every table and quoted number in `tables/`. The tcal product is built by `eigsep_data`'s commands `eigsep-build-tcal`, `eigsep-validate-tcal` and `eigsep-tcal-bracket-cost`, and the noise-source product by `eigsep-build-noise-source` (`eigsep_data.calibration`), each with the choices in its version's `config.json`. Christian's chain, which the script runs for the night's summary and the S11 cross-check, is `eigsep_data.calibration.d5_yfactor`.

The script reads files from the package checkouts `eigsep_cal`, `eigsep_data` and `eigsep_sim`, and stamps each with its commit. How it finds them depends on how the packages are installed:

- **Editable install** (`pip install -e`): from where `eigsep_data` is imported. Nothing to set; the code that runs is the checkout, so its commit is the code's commit.
- **Regular (fixed) install**: inside the top-level `eigsep` checkout this memo belongs to, which the script finds through git, also from a worktree. The installed code is a snapshot, so reinstall `eigsep_cal` and `eigsep_data` from their checkouts before running. If an installed copy differs from its checkout, its stamp below reads "(installed copy differs)" and the run is not reproducible from the listed commits.
- **Other layouts**: set `EIGSEP_SIBLINGS` to the directory that holds the package checkouts. The script stops with a message if it finds none.

Run it on a clean tree, with the memo source committed, or this repository's stamp ends in `-dirty`.

input	SHA-256 (first 16)
`marjum-2026-07/derived/tcal/v0004/solutions.npz`	`54d6d05dbe0e7ad2`
`marjum-2026-07/derived/tcal/v0003/validation.json`	`b5c3390d2ecbd4b3`
`marjum-2026-07/derived/noise_source/v0001/solutions.npz`	`baf14d314bb5d07d`
`marjum-2026-07/data/s11/cal_materials/rfn_sparameters_10_15.npz`	`a0ab5df768de2bd7`
`marjum-2026-07/data/s11/cal_materials/lab/20261015_131917_vna_data.npz`	`1f78b1e2256fea76`
`marjum-2026-07/data/s11/cal_materials/switch_sparams.npz`	`8b4418f07187c965`
`eigsep_cal/src/eigsep_cal/data/calibrated_internal_osl2026.npz`	`832ba837e98017f5`
`marjum-2026-07/data/s11/cal_materials/vna_internal_osl_20260629T090825Z.npz`	`3a0e93ff130e4287`
`eigsep_data/hfss_beam_maps/bowtie_beam.npz`	`c4046f5e9332cd57`
`eigsep_sim/src/eigsep_sim/data/S11_eigsep_bowtie_v000.npz`	`a8abe8bc6ad9a9d8`
`eigsep_data/src/eigsep_data/calibration/d5_yfactor.py`	`0ed1248d7f4e6ee4`
40 field S11 files (checked against `manifest.csv`)	all match
292 correlator files of the 16/17 Jul night (digest of SHA-256s)	`a2f684c6cd25bd16`
72 files of the 2026-10-08 bench run (checked against the noise_source manifest; digest)	`68f7ace133aab336`
8 raw switch-path captures, `cal_materials/lab/switch_paths` (digest)	`b83c80687daf5a80`
eigsep commit	`e291641`
eigsep_cal commit	`1362bdb`
eigsep_data commit	`466c0d1`
eigsep_sim commit	`d050c83`
tcal v0004 built with eigsep_data / eigsep_cal	`e1da3c5 / 1362bdb`
noise_source v0001 built with eigsep_data / eigsep_cal	`e1da3c5 / 1362bdb`
generated	2026-10-24 00:18 UTC

B The S11 calibration, step by step

The intermediate results of §4.3: the lab characterization of the internal standards (Figure 13), and one field sweep of the antenna and one of the receiver at every stage of the field calibration (Figure 14). The procedure and its code are in `sparam_characterization.tex` and `EIGSEP.github.io/notebooks/characterizing_system_2026.ipynb`; the figures here are drawn by `make_products.py` from the tracked captures.

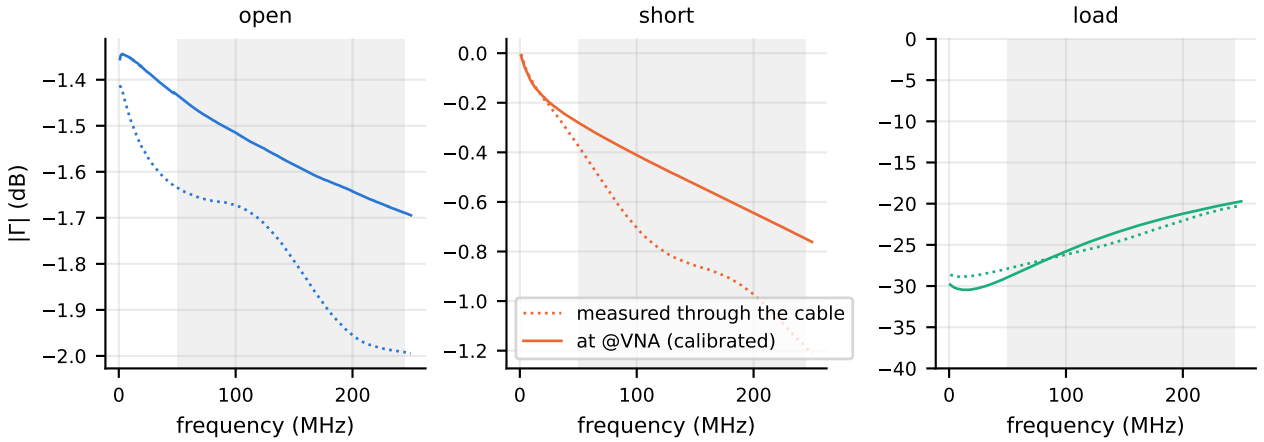


Figure 13: The switch matrix's internal open, short and load, as the field sweeps use them. Dotted: measured through the VNA cable; solid: at @VNA, after removing the VNA-and-cable network solved from S911T standards on the cable's free end (lab, 2026-09-12). Each field sweep solves its VNA network against these values; rederived from the capture they match `eigsep_cal`'s packaged file to 9×10^{-16}.

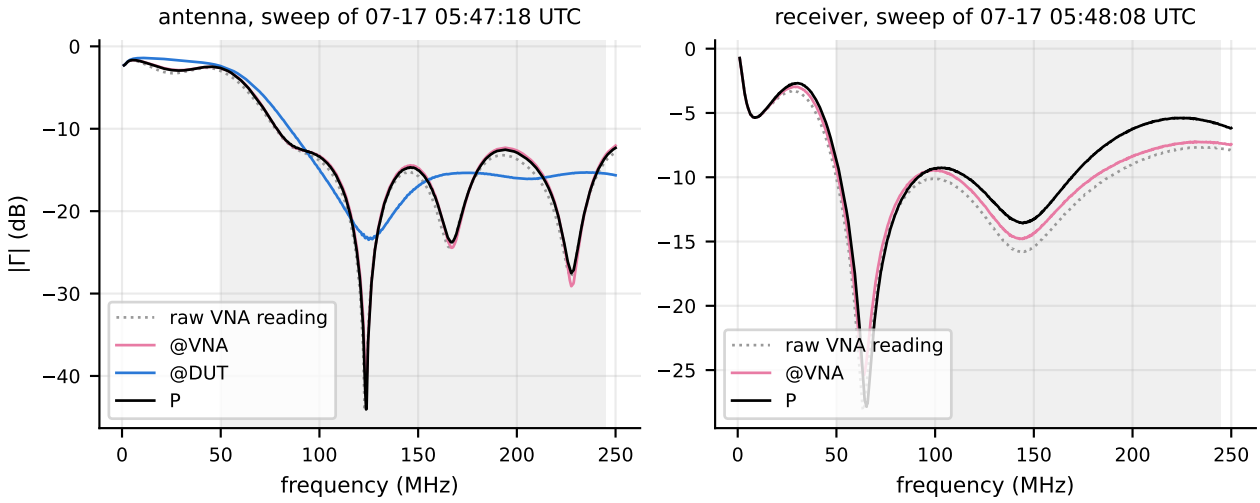


Figure 14: One field sweep of the antenna (left, 07-17 05:47 UTC) and one of the receiver (right, 07-17 05:48 UTC) on the 16/17 July night, at each stage of `eigsep_cal.S11`. Antenna: the raw VNA reading; at @VNA, after removing the VNA network solved from that sweep's own internal standards; at @DUT, after removing the `VNAANT` path; at P, after adding the `RFANT` path. Receiver: raw, at @VNA, and at P after removing `VNARF`.